

 UTM UNIVERSITI TEKNOLOGI MALAYSIA	Software Engineering Dept Faculty of Computing Universiti Teknologi Malaysia
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SECJ3483 - WEB TECHNOLOGY

GROUP PROJECT - UTM MarketHub

Report

Prepared by:

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1. Introduction

UTM MarketHub is a campus-oriented online marketplace developed to facilitate the buying and selling of second-hand items within the Universiti Teknologi Malaysia community. The platform provides a centralised web-based environment in which students can browse available products, publish their own listings, manage offers, and interact with other users through a structured marketplace process.

The system was designed as a single-page web application to provide users with a responsive and seamless browsing experience. Its frontend was developed using Vue.js, while the backend was implemented using the PHP Slim Framework as a RESTful Application Programming Interface. The platform also uses a MySQL database to store and manage user information, product categories, marketplace listings, and offer records.

By combining modern frontend and backend technologies, UTM MarketHub aims to deliver a practical, secure, and user-friendly platform that addresses the specific buying and selling requirements of the university community.

2. Problem Background

University students frequently require affordable access to items such as textbooks, electronic devices, stationery, furniture, clothing, and other daily necessities. At the same time, students may possess unused items that remain in good condition and could be sold to other members of the campus community. A second-hand marketplace can therefore provide financial benefits to both buyers and sellers while extending the useful lifespan of existing products.

However, the buying and selling of second-hand items within the university environment are commonly conducted through informal communication channels, including social media posts, messaging applications, and separate student chat groups. Although these channels enable basic communication, marketplace information is often scattered across different platforms. As a result, potential buyers may find it difficult to locate suitable items, compare available listings, or determine whether a product remains available.

The absence of a dedicated marketplace platform also limits the ability of users to search and filter items according to relevant attributes such as category, condition, price, and listing status. Sellers may experience difficulty reaching the appropriate target audience, while buyers may overlook relevant products because marketplace posts are quickly replaced by newer messages.

Furthermore, informal buying and selling processes provide limited support for managing offers and listing ownership. Sellers may receive multiple offers through different channels, making it difficult to monitor, accept, or reject them systematically. Similarly, administrators do not have a dedicated interface for maintaining the product categories used within the marketplace. These limitations demonstrate the need for a centralised and structured campus marketplace system.

3. Proposed Solution

UTM MarketHub is proposed as a centralised campus marketplace that integrates product discovery, listing management, offer handling, and category administration within a single web application. The system is intended to provide a more organised and efficient alternative to marketplace activities conducted through fragmented social media and messaging channels.

The platform supports public marketplace browsing, allowing both guests and registered users to view available products. The landing page introduces the purpose of the platform and presents recent marketplace listings to help users discover newly published items. This page serves as the primary entry point to the system and provides users with an overview of the services offered by UTM MarketHub.

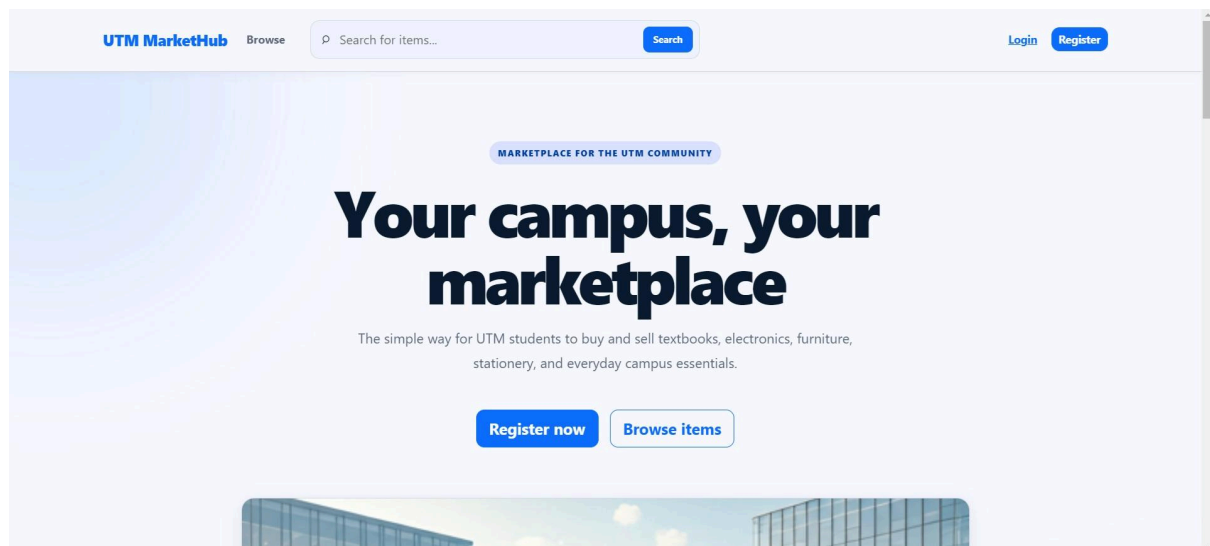


Figure 01: Home Page

The marketplace results page provides users with a comprehensive view of available product listings. Through this page, users can search for specific products, filter listings according to selected criteria, sort marketplace results, and compare available items. These functions improve product discoverability and allow users to identify suitable listings more efficiently.

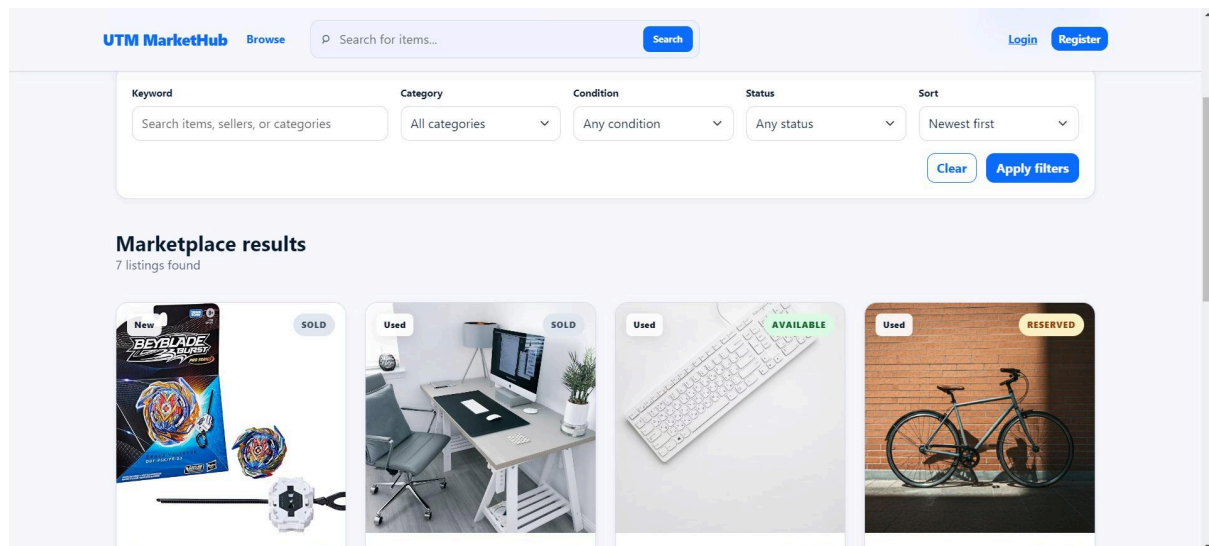


Figure 02: Market Results Page

Each marketplace listing is presented through a dedicated listing details page. This page displays essential product information, including the product title, description, price, condition, category, seller information, and listing availability. Registered users who own a listing are also provided with the appropriate controls to manage and update their product information.

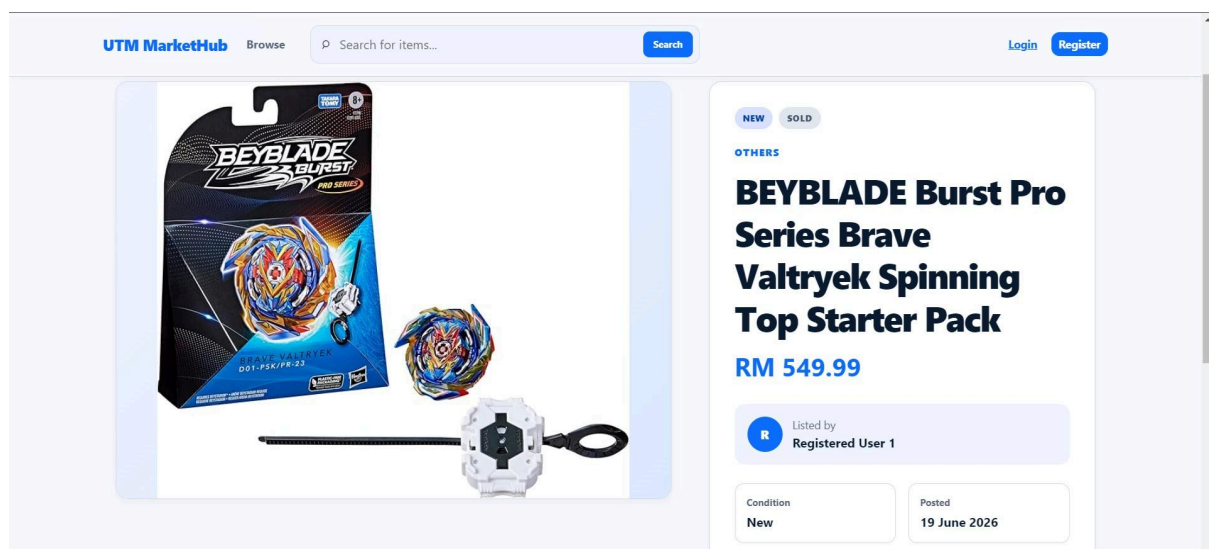


Figure 03: Listing Detail Page

The system includes a listing creation and editing interface that enables authenticated users to publish new products and modify their existing listings. Users are required to provide the relevant product information before submitting a listing. This structured approach ensures

that marketplace listings contain sufficient information to support informed purchasing decisions.

UTM MarketHub Browse My listings Offers Sell Search for items... Search Registered User 1 Logout

Edit your listing

Keep the item details and availability up to date.

LISTING INFORMATION

Describe your item clearly

Accurate details help other students understand what you are selling.

Item title

BEYBLADE Burst Pro Series Brave Valtreyk Spinning Top Starter Pack

Description

Top with die-cast metal parts for a heavy weight, rubber attack blades, and gradual performance tip change.

106 / 5000

PREVIEW

Your listing card



Figure 04: Create/Edit Listing Page

UTM MarketHub also incorporates an offer management module that facilitates communication and negotiation between buyers and sellers. Through this module, interested buyers can submit offers for available items, while sellers can review the offers associated with their listings. Sellers may subsequently accept or reject an offer based on their preferences. This process provides a clear and traceable mechanism for managing marketplace transactions.

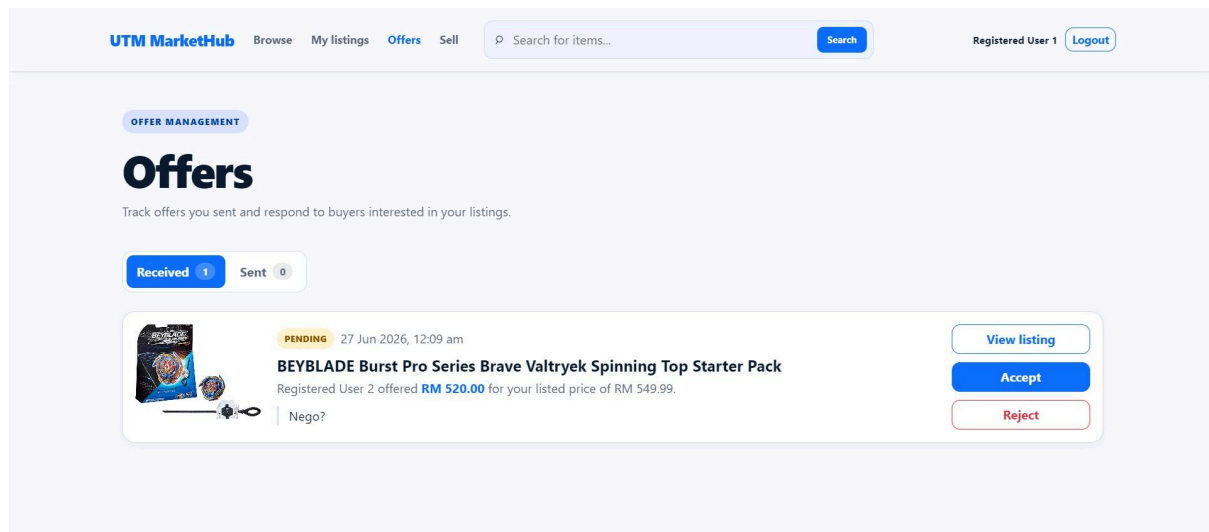


Figure 05: Offers Page

In addition to user-oriented marketplace functions, the system provides an administrative category management interface. This interface allows authorised administrators to create, view, update, and remove marketplace categories. Centralised category management ensures that listings are consistently organised and enables users to browse products according to clearly defined classifications.

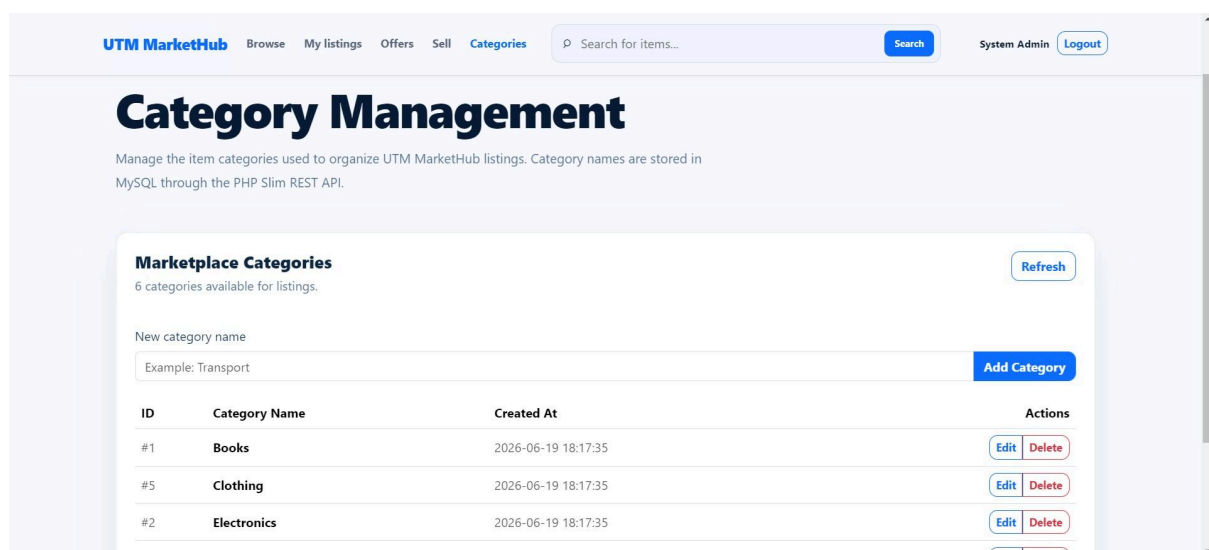


Figure 06: Admin Categories Page

From a functional perspective, the system implements secure user authentication using JSON Web Tokens. Authentication is required for protected operations such as creating listings, editing listings, submitting offers, and managing user-owned marketplace content. Guests are permitted to browse public listings, while registered users receive access to additional marketplace functions.

The system also implements Create, Read, Update, and Delete operations for marketplace listings. Registered users can create and manage their own listings, while appropriate authorisation controls prevent users from modifying content owned by other individuals. Offer workflows are similarly restricted according to user roles and listing ownership.

Administrative operations, particularly category management, are restricted to users with administrator privileges. This role-based access control mechanism protects sensitive system functions and ensures that only authorised individuals can modify the marketplace structure. All persistent information, including user accounts, categories, listings, and offers, is stored in a MySQL relational database.

4. Tech Stack

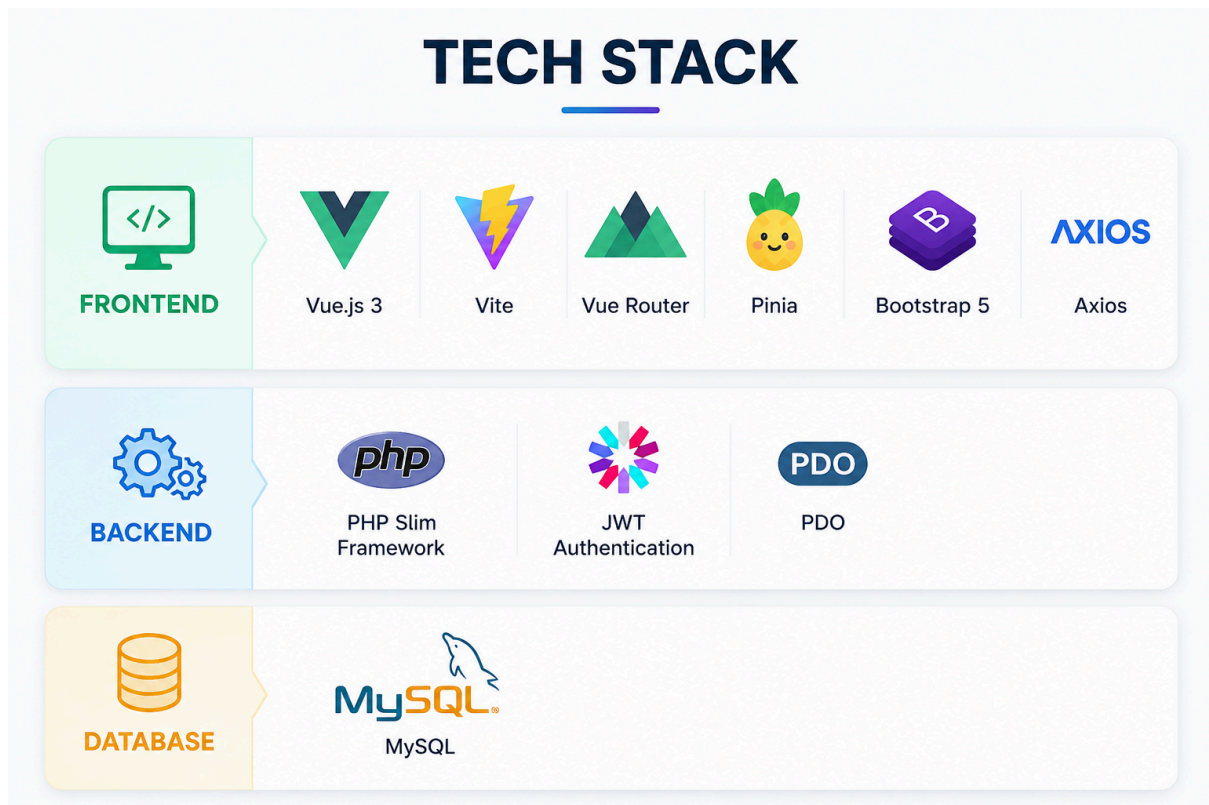


Figure 07: Tech Stack

5. Impact

UTM MarketHub provides practical value to the university community by making second-hand buying and selling more accessible, organised, and efficient. Instead of relying on informal and fragmented communication channels, students are provided with a dedicated digital platform that improves the visibility of available products and offers greater control over marketplace activities.

The platform enables sellers to present their items in a structured format and reach users who are actively searching for second-hand products. Buyers benefit from the ability to search, filter, sort, and compare listings based on their specific requirements. The offer management process further improves the interaction between buyers and sellers by providing a systematic method for submitting and responding to purchase offers.

The project also supports sustainable and community-oriented outcomes:

- SDG 12: Responsible Consumption and Production by encouraging the reuse and recirculation of goods such as books, electronics, and furniture instead of buying new items unnecessarily.
- SDG 11: Sustainable Cities and Communities by promoting a more resource-sharing campus environment and reducing waste within the university community.
- SDG 8: Decent Work and Economic Growth by helping students save money and create small peer-to-peer selling opportunities.

6. Conclusion

UTM MarketHub addresses an identifiable need within the UTM community by providing a dedicated and structured platform for campus-based second-hand trading. The system consolidates marketplace activities that would otherwise be conducted across disconnected social media posts, messaging applications, and student groups.

Through the integration of marketplace browsing, listing management, offer handling, category administration, and secure user authentication, the platform improves the overall efficiency and organisation of the buying and selling process. Its use of Vue.js, the PHP Slim Framework, MySQL, RESTful API principles, and JSON Web Tokens demonstrates the practical application of modern web development technologies and architectural concepts.

Overall, UTM MarketHub represents both a functional web application and a meaningful contribution towards a more sustainable, connected, and resource-efficient university environment. The platform demonstrates how web technology can be applied to address real-world campus challenges while supporting responsible consumption, community engagement, and student-oriented economic activity.

7. Github Commit Summary

No.	Date	Contributor	Summary
1	Jun 27, 2026	Benjamin	Added system screenshots for report documentation and visual presentation of the completed features.
2	Jun 26, 2026	Tan Zheng Yu	Updated and expanded the project report draft with additional content.
3	Jun 20, 2026	Tan Zhen Li	Implemented the offer management module, including buyer and seller actions such as creating, updating, accepting, rejecting, and cancelling offers.
4	Jun 19, 2026	Benjamin	Added functionality for uploading local images when creating or managing marketplace listings.
5	Jun 19, 2026	Benjamin	Developed the marketplace listing module, allowing users to create and manage product listings.
6	Jun 15, 2026	Teow Zi Xian	Implemented the admin category management module for maintaining marketplace categories.
7	Jun 9, 2026	Tan Zheng Yu	Added the authentication module, landing page, and database setup tools for the project.
8	Jun 9, 2026	Tan Zheng Yu	Created the initial project structure using Vue for the frontend and PHP Slim for the backend.